

Software Requirements Specification (SRS): Banking Management System

1. Project Overview

The Banking Management System is a comprehensive digital platform designed to provide secure, efficient, and intelligent financial services to customers while providing robust management and oversight capabilities for bank staff and administrators.

2. Executive Summary

The system aims to centralize banking operations including account management, automated transaction processing, credit facilities, and AI-driven financial advisory tools. The core focus is on security, user autonomy in financial management, and operational efficiency for internal staff.

3. Stakeholders

- **Customers:** End-users accessing banking services.
- **Bank Employees:** Staff responsible for verifying and approving loan applications.
- **Admins:** Users responsible for system monitoring, user management, and operational oversight.
- **Compliance/Audit Team:** Stakeholders monitoring fraud detection logs.

4. Target Users

- **Retail Banking Customers:** Individuals seeking to manage funds, pay bills, and apply for loans.
- **Operational Staff:** Employees tasked with loan underwriting and account verification.
- **System Administrators:** Personnel managing system health, security configurations, and user access.

5. Functional Requirements

- **Account Management:** Users shall be able to create accounts, update profiles, and view balances.
- **Transaction Processing:** Users shall be able to transfer funds and pay bills.
- **Loan Management:** Customers shall be able to apply for loans; employees shall be able to review and approve/reject applications.

- **AI Financial Insights:** The system shall provide spending analysis and budgeting recommendations.
- **Fraud Detection:** The system shall monitor transactions for suspicious activity and trigger alerts.
- **Security:** The system shall enforce Multi-Factor Authentication (MFA) for all user roles.
- **Admin Dashboard:** Admins shall have access to system logs, user activity reports, and fraud status indicators.

6. Non-Functional Requirements

- **Security:** All financial data must be encrypted in transit and at rest.
- **Availability:** The system shall maintain 99.9% uptime.
- **Performance:** Transactions must be processed within 2 seconds under normal load.
- **Scalability:** The system shall support concurrent growth in user base and transaction volume.
- **Auditability:** All actions performed by employees or admins must be logged with a timestamp and user ID.

7. User Stories

- **As a Customer,** I want to set a monthly budget so that I can receive alerts when I am close to my limit.
- **As a Customer,** I want to transfer money to another account so that I can pay others easily.
- **As an Employee,** I want to view a list of pending loan applications so that I can process them in order of priority.
- **As an Admin,** I want to view a dashboard of flagged fraudulent transactions so that I can mitigate security risks.

8. Acceptance Criteria

- **MFA:** User is unable to access the account without entering a one-time code sent to their registered device.
- **Loan Approval:** A loan application status changes to "Approved" only after an authorized employee performs the action.
- **Budgeting:** System generates a report of spending vs. defined budget categories at the end of every month.

9. Business Rules

- Loans cannot be approved without a credit score check completion.
- Daily transfer limits are enforced based on account type.
- Any transaction exceeding \$10,000 must trigger an automatic fraud review flag.

10. Assumptions

- The system will integrate with a third-party credit bureau service.
- Users have access to an internet-connected device with a modern browser/app capability.
- Legal requirements for regional banking are provided by the client.

11. Constraints

- Must comply with GDPR and local financial regulatory standards.
- Limited to standard currency transactions as defined by the bank.

12. Risks

- **Security Breach:** Exposure of sensitive financial data (Mitigation: Regular penetration testing).
- **False Positives in Fraud Detection:** Legitimate transactions blocked (Mitigation: Fine-tuning AI sensitivity).
- **System Downtime:** Disruption of financial services (Mitigation: Redundant infrastructure).

13. Out of Scope

- Cryptocurrency trading and wallet management.
- Physical hardware (ATMs, branch scanners).
- Stock trading and investment portfolios.
- Integration with legacy mainframe banking systems (Phase 1).

14. Open Questions

- What is the specific list of supported bill-pay providers?
- Should the AI insight model rely on custom algorithms or third-party integration?

- What are the specific recovery procedures for lost MFA credentials?